



Multilevel and Granular L2 Pronunciation Assessment Using Stress-Based Suprasegmental Features and Proficiency Adaptation

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Abstract. The joint modeling of mispronunciation detection and diagnosis (MDD) and automatic pronunciation assessment (APA) in computer-assisted pronunciation training (CAPT) systems has been proven effective. While existing approaches (e.g., HMamba) employ self-supervised learning (SSL) to extract rich speech representations incorporating prosodic features like phoneme duration and silent segments, they still face two key challenges: On one hand, SSL representations lack targeted modeling of crucial suprasegmental features such as stress; on the other hand, directly joint-training segmental features (local phoneme accuracy) and suprasegmental features (global prosodic patterns) leads to performance conflicts due to their differing granularities of focus. To address these issues, this paper proposes a stress-enhanced framework. First, we explicitly model word-level and sentence-level stress features based on vowel formants (F1/F2) and spectral balance. Subsequently, we employ a proficiency-aware attention matching mechanism that adaptively adjusts fusion weights between segmental and suprasegmental features according to learners' second-language proficiency.

Keywords: MDD · APA · CAPT · proficiency · Mamba

1 Introduction

Mispronunciation Detection and Diagnosis (MDD) and Automatic Pronunciation Assessment (APA) have traditionally been treated as separate tasks in Computer-Assisted Pronunciation Training (CAPT) systems. While MDD has achieved significant progress with state-of-the-art models reaching 63% F1-score on the L2-ARCTIC dataset [14], these approaches primarily focus on segmental-level features, neglecting crucial suprasegmental feedback that is equally important for second language (L2) learners.

The field of APA witnessed a major advancement with the release of the Speechocean762 dataset [18] and the introduction of GOPT [9], which established a new baseline. Subsequent studies [4, 17] have mainly focused on architectural modifications of GOPT with limited improvements. Recent works [6, 12, 15]

attempted to incorporate different ASR models for multi-level assessment but achieved marginal gains.

The potential of joint MDD-APA modeling was first demonstrated by [13] on proprietary data, while HMamba [3] recently extended this approach to Speechocean762 using Mamba architecture for computational efficiency, showing improvements across all assessment levels. However, English being a stress-timed language (e.g., “REcord” vs. “reCORD” where stress determines lexical meaning), the evaluation of word stress remains understudied. Current stress-level Pearson Correlation Coefficient (PCC) scores show room for improvement, and sentence-level completeness scores stagnate around 0.2, significantly lagging behind other metrics.

Second language (L2) proficiency significantly impacts pronunciation assessment outcomes, with advanced learners demonstrating better segmental accuracy and stress patterns compared to beginners, as shown in longitudinal studies of L2 English learners [11]. The correlation between vocabulary size and pronunciation scores ($r = 0.62$, $p < 0.01$) particularly highlights the linguistic threshold effect in computer-assisted pronunciation training systems [5]. Recent work further confirms that incorporating L2 proficiency levels as a conditional variable improves automatic assessment accuracy by 12.7% in state-of-the-art models [10].

Our goal is to develop a CAPT framework capable of effectively modeling both segmental and suprasegmental information, with particular focus on optimizing word stress assessment through proficiency-aware attention mechanisms. Current approaches suffer from limited feature diversity, primarily relying on goodness of pronunciation (GOP) features that only capture phoneme-level information. While self-supervised learning (SSL) models like HuBERT and Wav2Vec can extract richer representations through sophisticated data selection and training objectives, their features remain overly generic for pronunciation assessment. Although recent work (e.g., HMamba) has incorporated pause duration and phoneme length to better model suprasegmental features, these features lack specificity for stress detection. To address this, we propose targeted stress features including formant trajectories, spectral balance, and continuous pitch contours, which have been proven effective for stress analysis.

A critical challenge emerges in jointly modeling segmental and suprasegmental features - our experiments show that direct feature fusion degrades phoneme recognition performance ($F1 = 0.5649$), highlighting the need to preserve phonemic information while enhancing prosodic modeling. Our solution introduces a novel hierarchical attention matching mechanism (inspired by AAAI 2025 work [2]) that operates at both phoneme and word levels, adaptively combining English proficiency indicators with suprasegmental features. This approach achieves significant improvements in word stress PCC scores while maintaining phoneme recognition accuracy, with continuous pitch features additionally contributing to utterance-level completeness assessment.

2 Method

2.1 Feature Representation

Features Extraction for Segmental and Phonological Modeling. The Goodness of Pronunciation (GOP) metric, adapted from [16], serves as a core feature in our network. The GOP score is computed as:

$$G(p) = \frac{1}{\tau(p)} \ln \left| \frac{\mathcal{L}(p|o^p)}{\sum_{q \in \mathcal{Q}} \mathcal{L}(q|o^q)} \right| \quad (1)$$

$\tau(p)$: Duration of phoneme p (in frames)

$\mathcal{L}(p|o^p)$: Likelihood of phoneme p given observation o^p

$\mathcal{Q}$: Candidate phoneme set

The GOP feature extraction is implemented through our modified Kaldi recipe `gop_speechocean762`, which is adapted from Kaldi’s standard GOP implementation. This pipeline utilizes a hybrid DNNHMM acoustic model trained on 80 dimensional logmel filterbank features. The system performs forced alignment at the phoneme level and applies speaker-level cepstral mean and variance normalization (CMVN) to ensure robustness across different speakers. The GOP scores are then computed based on the phoneme posterior probabilities obtained from this alignment process. We enhance phoneme representations with both relative and absolute positional embeddings. Relative positions are encoded using standard BIES tags (Beginning/Inside/End/Single), while absolute positions employ learned embeddings indexed by phoneme order in the utterance.

Acoustic Correlates of Stress. The acoustic feature extraction pipeline combines formant and spectral characteristics [1] into a 6D vector. Formant features $\mathbf{f}_{\text{formant}} = [F1, F2] \in \mathbb{R}^2$ are sampled at vowel midpoints $t_{\text{mid}} = (t_{\text{start}} + t_{\text{end}})/2$ using Parselmouth’s Burg method with 4-formant tracking. Spectral features $\mathbf{f}_{\text{spectral}} \in \mathbb{R}^4$ capture band energies $E_{a:b} = 10 \log_{10}(\mathbb{E}[|S(f)|^2])$ across four frequency ranges (0–500 Hz, 500–1000 Hz, 1000–2000 Hz, 2000–4000 Hz) computed via 2048-point STFT. The final representation $\mathbf{f}_{\text{acoustic}} = \mathbf{f}_{\text{formant}} \oplus \mathbf{f}_{\text{spectral}} \in \mathbb{R}^6$ concatenates both feature types. The pitch contour is extracted using PyWORLD’s DIO algorithm with 20ms frame length and 20ms hop size, clamped to 100–600 Hz range and converted to 256 Mel-scaled bins. Phoneme durations are obtained through forced alignment, while energy features follow [7]’s RMS energy statistics. Formant and spectral features complete the 6D acoustic vector. These features are combined with segmental features as inputs.

2.2 Proficiency Embeddings

As noted in [18], the Speechocean762 dataset was originally constructed by categorizing speakers into three proficiency levels (high, medium, low) during train-test splitting, though these explicit proficiency labels were not publicly released. In our work, we derive comparable proficiency estimates by:



Fig. 1. Comparison of Test vs Train Set by Proficiency in spo762 dataset.

1. Calculating per-speaker mean scores across all utterances
2. Dividing speakers into tertiles (low/medium/high) based on score distribution
3. Validating that our grouping closely matches [18]’s original stratification(see Fig. 1)

These discrete levels are then processed through:

$$\mathbf{p} = \text{nn.Embedding}(3, d) \in \mathbb{R}^{L \times d} \quad (2)$$

where L is sequence length and d the embedding dimension, enabling seamless integration with acoustic features via ABM.

2.3 Feature Fusion with ABM Module

Inspired by the Attention-Based Matching (ABM) mechanism in [2], which was originally designed to address frequency distribution discrepancies between source (annotated data) and target (new categories) domains in cross-domain medical image segmentation through frequency-aware attention alignment, we adapt this approach for pronunciation assessment. At the phoneme level, we employ ABM to effectively integrate learner proficiency, enabling the model to perceive pronunciation deviations across speakers of different skill levels. For word- and utterance-level processing, building upon established findings that demonstrate the correlation between suprasegmental feature acquisition and age-related proficiency levels [8], we utilize ABM to combine suprasegmental features with proficiency embeddings. See Fig. 2)The detailed formulation is as follows:

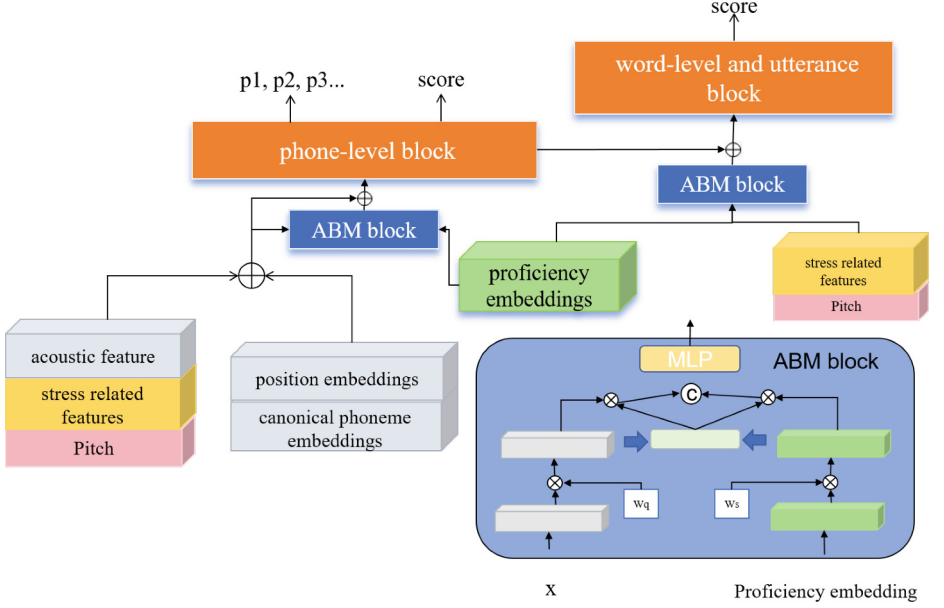


Fig. 2. The Proposed MDD&APA Framework with stress features, pitch, proficiency embeddings and ABM(attention-based matching) added.

1. **Linear Transformation:** Given proficiency embedding $\mathbf{s} \in \mathbb{R}^{L \times 128}$ (sequence length L , dimension 128) and query features:

$$\begin{cases} Q_{\text{phone}} = \mathbf{H} \in \mathbb{R}^{L \times d} & (\text{Phone-level ABM}) \\ Q_{\text{word}} = \text{MLP}(\text{concat}[\mathbf{d}, \mathbf{e}, \mathbf{st}, \mathbf{p}]) & (\text{Word-level ABM}) \end{cases} \quad (3)$$

where $\mathbf{d} \in \mathbb{R}^{L \times 1}$ (duration), $\mathbf{e} \in \mathbb{R}^{L \times 8}$ (energy), $\mathbf{st} \in \mathbb{R}^{L \times 6}$ (stress), and $\mathbf{p} \in \mathbb{R}^{L \times 1}$ (pitch) are supersegmental features.

2. **Projection:**

$$\begin{cases} \mathbf{F}_s = \mathbf{s} \mathbf{W}_s \\ \mathbf{F}_q = \mathbf{Q} \mathbf{W}_q \end{cases}, \quad \mathbf{W}_s, \mathbf{W}_q \in \mathbb{R}^{d \times d} \quad (4)$$

3. **Attention Computation:**

$$\mathbf{A}(\mathbf{F}_s, \mathbf{F}_q) = \sigma \left(\frac{\mathbf{F}_s \mathbf{F}_q^\top}{\|\mathbf{F}_s\| \|\mathbf{F}_q\|} \right) \in \mathbb{R}^{L \times L} \quad (5)$$

3 Experiments

3.1 Datasets

The Speechocean762 dataset [18] is an open-source resource designed for pronunciation assessment, consisting of 5,000 English utterances from 250 non-native

Mandarin speakers, split equally into training and test sets. The dataset includes rich annotations at multiple granularities: at the utterance level, it provides scores for accuracy, fluency, completeness, prosody, and a total score (0–10); at the word level, it offers accuracy, stress, and total score (0–10); and at the phoneme level, each phoneme is annotated with an accuracy score (0–2). All scores are provided by five experts, resulting in a total of 8 distinct labels. The utterance and word-level scores to match the 0–2 range of the phoneme scores in this study. In the MDD task, the dataset includes 46 phones, with mispronunciations annotated using [unk] for unknown phones and [DEL] for deletion errors, allowing alignment with canonical phones. Detailed information like speaker characteristics, duration and Native Language(L1) and are provided in Table 1.

Table 1. Statistical Summary of the SpeechOcean762 Dataset: Speakers, Duration, Utterances(Utt.) and Language Distribution

	Train	Test
Speakers	125	125
Duration(h)	2.7	2.8
Utt.	2500	2500
Words	15849	15967
phones	47076	47369
L1	Chinese	

3.2 Experimental Settings

Our framework builds upon **Hmamba**¹, an open-source state-of-the-art joint architecture for Automatic Pronunciation Assessment (APA) and Mispronunciation Detection and Diagnosis (MDD) [3].

The model’s optimizers are Adam and a tri-phase learning rate scheduler. The initial learning rate was set to 2e-3, except for the utterance-level APA module, where it was set to 9e-5. To address the pronounced class imbalance in pronunciation assessment (where correct pronunciations significantly outnumber mispronunciations), we adapt a **Decoupled Cross-Entropy Loss (deXent)** that explicitly balances positive and negative samples:

$$\mathcal{L}_{MDD} = \underbrace{\mathcal{L}_{Xent}^{hit}}_{\text{Correct pronunciations}} + \underbrace{\left(\frac{\mu^h}{\mu^m}\right)^\alpha \mathcal{L}_{Xent}^{mis}}_{\text{Re-weighted mispronunciations}} \quad (6)$$

where:

¹ <https://github.com/Fuann/hmamba>.

- $\mathcal{L}_{Xent}^{hit}$ and $\mathcal{L}_{Xent}^{mis}$ are cross-entropy losses for correctly pronounced ($\mathcal{H}$) and mispronounced ($\mathcal{M}$) segments respectively
- μ^m , μ^h denote mispronunciation and correct pronunciation frequencies in training data
- α controls the re-weighting intensity (empirically set to 0.5)

These settings are consistent with those used in HMAMBA [3]. The evaluation metrics include the Pearson Correlation Coefficient (PCC) and Mean Squared Error (MSE) for the APA task for Score re-scaling for utterance/word levels does not affect PCC/MSE. For the MDD task, we used precision, recall, F1-score, and phone error rate (PER), in line with prior studies. To ensure result validity, we performed 5 independent trials for each experiment, running 20 epochs with different seeds. The reported metrics are the average across these trials.

Table 2. Overall Performance of APA³

Model	Utterance Score					Word Score			Phone Score	
	Prosodic $\uparrow$	Fluency $\uparrow$	Complete $\uparrow$	Accuracy $\uparrow$	Total $\uparrow$	Accuracy $\uparrow$	Stress $\uparrow$	Total $\uparrow$	MSE $\downarrow$	Total $\uparrow$
GOPT	0.760	0.753	0.155	0.714	0.742	0.533	0.291	0.549	0.085	0.612
HMamba	0.835	0.842	0.172	0.805	0.825	0.708	0.313	0.716	0.062	0.736
Our Method										
1. +Stress	0.843	0.849	0.234	0.809	0.830	0.707	0.363	0.717	0.062	0.735
2. +phone-ABM (on 1)	0.857	0.854	0.161	0.828	0.851	0.701	0.358	0.711	0.063	0.731
3. +Word-ABM (on 2)	0.857	0.849	0.222	0.824	0.850	0.703	0.337	0.713	0.061	0.742
4. +Pitch (on 3)	0.849	0.841	0.255	0.822	0.849	0.690	0.362	0.702	0.064	0.726
5. +prosodyEmb (on 4)	0.862	0.862	0.262	0.823	0.847	0.694	0.362	0.703	0.066	0.720

4 Experimental Results and Discussion

4.1 ABM Placement Matters: Feature Fusion Strategies Analyzed

As shown in Table 2, compared to the GOPT framework [9] using GOP as acoustic features and the HMAMBA framework [3] incorporating SSL features with GOP, duration, and energy inputs, our approach demonstrates significant improvements in suprasegmental metrics at both word and utterance levels through the explicit incorporation of targeted stress features (formants and spectral balance) and continuous pitch contours. The experimental results shows that the specifically designed stress features (formants + spectral balance) exhibit significant advantages in word-level stress assessment, which stems from three underlying mechanisms. Firstly, from an acoustic perspective, the perception of English stress primarily relies on vowel formant structures (F1/F2) and energy distribution in specific frequency bands (e.g., 500-1000 Hz), fine-grained features that are overlooked by the HMM likelihood used in traditional GOP. Secondly, in terms of modeling strategy, the pre-concatenation of stress features before word-level modeling enables the neural network’s lower layers to directly process

raw spectral details. Lastly, from an articulatory physiology standpoint, our feature design accurately captures the vocal cord tension changes (reflected in F1 elevation) and high-frequency energy enhancement (energy difference $>3\text{dB}$ in the 2000-4000 Hz band) that accompany stressed syllables.

In contrast, continuous pitch contours demonstrate stronger explanatory power for utterance-level completeness assessment. The underlying mechanism lies in the linguistic-cognitive perspective: the perception of utterance completeness is closely related to the phenomenon of pitch reset—the pitch rise at the beginning of a sentence marks the start of a turn, while the pitch convergence at the end signals semantic completion. Only continuous pitch features can fully encode these critical prosodic events. Notably, the optimal performance of Configuration 5 (prosodyEmb + proficiency features) on prosodic metrics essentially reflects the synergistic effect of multi-level representation fusion. The word-ABM mechanism dynamically adjusts the interaction intensity between acoustic features and proficiency embeddings through attention weights, focusing on spectral detail modeling at the word level while enhancing intonation pattern integration at the utterance level. This adaptive fusion strategy surpasses the limitations of single-feature combinations (see Sect. 4.2 for the architecture diagram). These findings provide empirical evidence and theoretical guidance for feature engineering and model design in pronunciation assessment systems.

4.2 Effectiveness of Different Prosody-Related Features

To investigate how different feature configurations can balance the performance between suprasegmental scoring and segmental-level MDD (Mispronunciation Detection and Diagnosis), we selected several key suprasegmental feature-based scores, along with the F1 score and PER (Phone Error Rate) from the segment-focused MDD system, as our evaluation metrics.

As shown in Table 3, the first configuration, prosodyEmb, denotes a concatenation of 1D duration, 7D energy, 6D stress (formats and Spectral), and 1D pitch into a unified embedding. Among these, duration and energy are features adopted from the Hmamba framework, while stress and pitch are newly introduced in this work. The prosodyEmb is integrated with the proficiency embedding within the sentence-level ABM module, providing informative suprasegmental representations at both the word and sentence levels.

We also conducted a series of ablation studies to assess the contribution of each prosodic feature. When duration was removed, a notable drop in completeness was observed. Further removing energy—leaving only stress and pitch—led to a configuration that not only maintained high performance but also achieved a good balance across metrics, demonstrating the effectiveness of the introduced features and we speculate that, compared to RMS energy, spectral balance is less influenced by changes in the speaker’s emotions and context, providing more stable speech features that help the model assess speech completeness more accurately. In some cases, these two features may offer redundant or conflicting information, increasing the difficulty for the model to integrate them. Removing one of the features may help reduce model complexity and improve performance. In

contrast, using only the stress feature (i.e., removing raw duration, energy, and pitch) resulted in the highest PCC (Pearson Correlation Coefficient) for stress, but a significant degradation in other metrics. This suggests that continuous pitch representations also contribute meaningfully to fluency, completeness, and even segmental-level modeling.

Table 3. Impact of Supersegmental Features on Supersegmental Score and MDD Performance

Model	Supersegmental Score			MDD	
	Stress	Fluency	Complete	F1 ↑	PER ↓
prosodyEmb	0.362	0.862	0.262	63.05%	3.16
–dur	0.362	0.857	0.197	60.08%	3.72
–dur-energy	0.372	0.857	0.314	63.92%	3.09
–raw(d,e)-pitch	0.376	0.857	0.197	60.08%	3.72

5 Conclusion

This study significantly enhances the performance of Computer-Assisted Pronunciation Training (CAPT) systems by explicitly incorporating stress-related features while accounting for speakers’ L2 proficiency levels. First, targeted stress modeling using formant trajectories and spectral balance features maintains phoneme recognition accuracy (F1=63.92%) while improving word stress assessment performance to a peak PCC of 3.76%. Second, the framework integrates learner proficiency as a key feature and employs an attentive matching mechanism to hierarchically fuse segmental and suprasegmental features, achieving consistent improvements in utterance-level suprasegmental scoring. Experimental results demonstrate that our hybrid approach combining engineered acoustic features with learned representations significantly outperforms purely self-supervised learning methods, particularly for supersegmental modeling. While currently optimized for English, the modular architecture enables future extensions including cross-linguistic formant normalization and tone-stress interaction modeling. Given stress’s critical role in English lexical distinction, subsequent research could develop more granular stress assessment metrics specifically targeting L2 learners’ stress production patterns.

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